

ISyE 3770 HW8

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Problem 8.1.1

Part A

Given the margin of error is $2.14 \frac{\sigma}{\sqrt{n}}$, we easily see $z = 2.14$.

Using a Z-Table, we find the following:

$$P(Z \leq 2.14) \approx 0.9838$$

$$P(Z \leq -2.14) \approx 1 - 0.9838 = 0.0162$$

$$\text{Confidence Level} = 0.9838 - 0.0162 = 0.9676 \text{ or } 96.76$$

Part B

We observe a Z-Score of 2.49.

Using a Z-Table, we find the following:

$$P(Z \leq 2.49) \approx 0.9934$$

$$P(Z \leq -2.49) \approx 1 - 0.9934 = 0.0066$$

$$\text{Confidence Level} = 0.9934 - 0.0066 = 0.9868 \text{ or } 98.68$$

Part C

We observe a Z-Score of 1.85.

Using a Z-Table, we find the following:

$$P(Z \leq 1.85) \approx 0.9678$$

$$P(Z \leq -1.85) \approx 1 - 0.9678 = 0.0322$$

$$\text{Confidence Level} = 0.9678 - 0.0322 = 0.9356 \text{ or } 93.56$$

Part D

We recognize this is a one-sided interval. We observe a Z-Score of 2.00.

Using a Z-Table, we find the following:

$$P(Z \leq 2.00) \approx 0.9772$$

$$\text{Confidence Level} = P(\mu \leq \bar{x} + 2.00 \frac{\sigma}{\sqrt{n}}) = P(Z \leq 2.00) = 0.9772 \text{ or } 97.72$$

Part E

We recognize this is a one-sided interval. We observe a Z-Score of 1.96.

Using a Z-Table, we find the following:

$$P(Z \leq 1.96) \approx 0.9750$$

$$\text{Confidence Level} = P(\mu \leq \bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}) = P(Z \leq 1.96) = 0.9750 \text{ or } 97.50$$

Problem 8.1.5**Part A**

A higher confidence interval would take longer. Therefore, we find that a 99% CI would be longer than a 95% CI since the interval must be wider to account for greater variability in the sample data.

Part B

The statement is not correct. For a CI X , we find X of samples of those intervals would contain the true mean μ . Furthermore, we find μ is not a random variable, so it is not within a specific interval. The provided statement treats μ as a random variable.

Part C

We find the statement to be correct. The statement correctly interprets confidence intervals, as outlined in Part B. Out of 1000 confidence intervals, we would expect ≈ 950 to include μ , which we find to be consistent with the CI.

Problem 8.1.7**Part A**

We find $Z_{0.005} \approx 2.5758$.

Therefore, we calculate standard error to be:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.001}{\sqrt{15}} = 0.0002582mm$$

We compute the margin of error to be:

$$ME = Z_{0.005}(SE) = 2.5758(0.0002582) = 0.000664$$

Therefore, we find the CI to be:

$$CI = (74.036 - 0.000664, 74.036 + 0.000664) = (74.0353mm, 74.0367mm)$$

Part B

We find $Z_{0.99} \approx 2.3263$. Therefore, we calculate standard error to be:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.001}{\sqrt{15}} = 0.0002582mm$$

We compute the margin of error to be:

$$ME = Z_{0.99}(SE) = 2.3263(0.0002582) = 0.000600$$

Therefore, we find the Lower Confidence Bound to be:

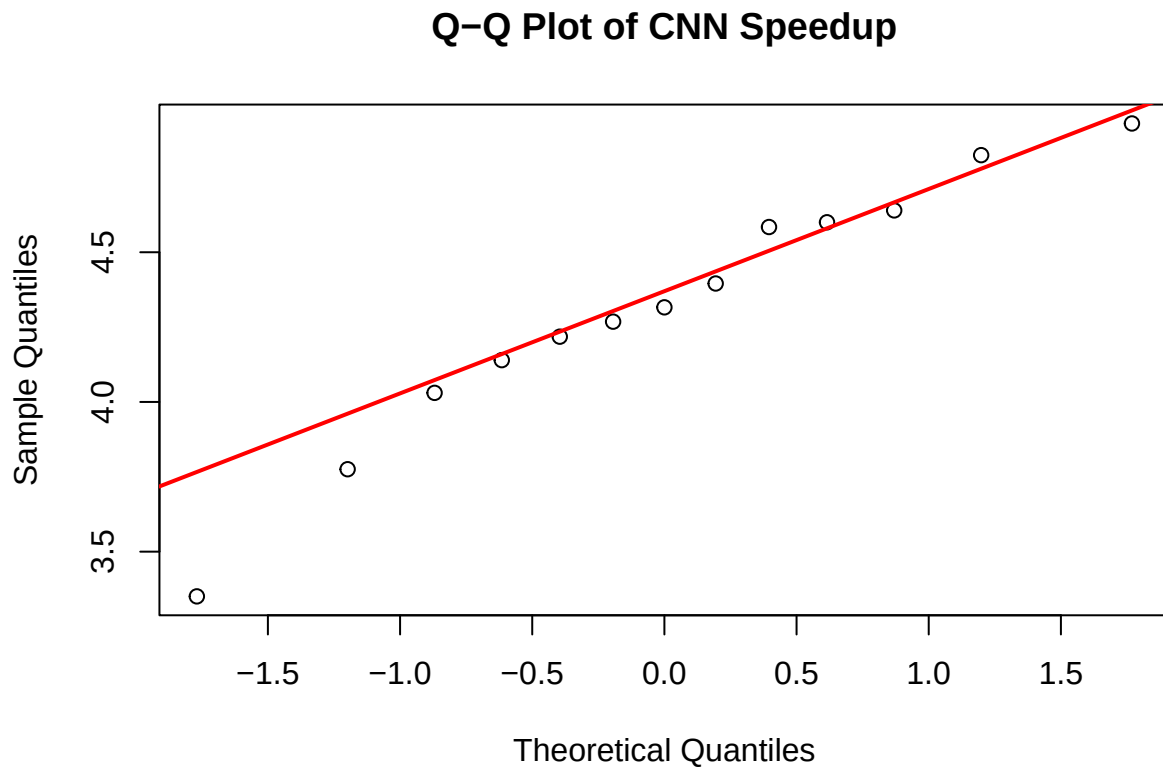
$$LCB = 74.036 - 0.000600 \approx 74.0354mm$$

We find the lower bound of the one-sided 99% confidence bound is slightly greater than that of the two-sided 99% confidence interval. We find this to be logical since the two-sided interval divides the significance level to each tail, thereby making it more conservative.

Problem 8.2.10

Part A

```
speedup <- c(3.775302, 3.350679, 4.217981, 4.030324, 4.639692,  
            4.139665, 4.395575, 4.824257, 4.268119, 4.584193,  
            4.930027, 4.315973, 4.600101)  
  
qqnorm(speedup, main="Q-Q Plot of CNN Speedup")  
qqline(speedup, col="red", lwd=2)
```



Since the data points in the Q-Q plot lie approximately on the red line, we can conclude the speedup of CNN is normally distributed.

Part B

We begin by calculating the sample mean:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{56.071888}{13} = 4.3132$$

We then calculate the sample standard deviation:

$$SST = \sum (x_i - \bar{x})^2 = 2.2478$$

Now, we find the variance to be:

$$S^2 = \frac{SST}{n-1} = \frac{2.2478}{12} = 0.1873$$

The standard deviation is trivial:

$$S = \sqrt{S^2} = 0.4328$$

Therefore, we can calculate the standard error:

$$SE = \frac{S}{\sqrt{n}} = \frac{0.4328}{\sqrt{13}} = 0.1201$$

We find the following:

$$t_{\alpha/2, n-1} = t_{0.025, 12} = 2.179$$

Therefore, we find the margin of error to be $2.179(0.1201) = 0.2616$.

Then, the lower limit is calculated by $4.3132 - 0.2616 = 4.0515$ and the upper bound is $4.3132 + 0.2616 = 4.5748$.

Part C

Using the values from part B, we simply find the following:

$$t_{\alpha, n-1} = t_{0.05, 12} = 1.782$$

Therefore, we find the margin of error to be $1.782(0.1201) = 0.2140$.

Finally, the lower bound is found by $4.3132 - 0.2140 = 4.0992$.

Problem 8.3.3

We begin by calculating the mean and standard deviation.

```
temperatures <- c(20.1, 20.7, 20.9, 21.3, 21.6, 21.7, 21.7, 23.3)
```

```
mean_temp <- mean(temperatures)
```

```
sd_temp <- sd(temperatures)
```

```
print(mean_temp)
```

```
## [1] 21.4125
```

```
print(sd_temp)
```

```
## [1] 0.9463275
```

For each temperature, we calculate the z-score.

```
z_scores <- (temperatures - mean_temp) / sd_temp
```

```
temperature_data <- data.frame(  
  Site = c("Harlquahala", "Laveen", "Maricopa", "Litchfield",  
           "Phoenix Encanto", "Phoenix Greenway", "Waddell", "Sky Harbor Airport"),
```

```

Temperature_C = temperatures,
Z_Score = round(z_scores, 3)
)

```

```

print(temperature_data)

```

```

##           Site Temperature_C Z_Score
## 1   Harlquahala      20.1  -1.387
## 2     Laveen      20.7  -0.753
## 3   Maricopa      20.9  -0.542
## 4   Litchfield      21.3  -0.119
## 5 Phoenix Encanto      21.6   0.198
## 6 Phoenix Greenway      21.7   0.304
## 7     Waddell      21.7   0.304
## 8 Sky Harbor Airport      23.3   1.995

```

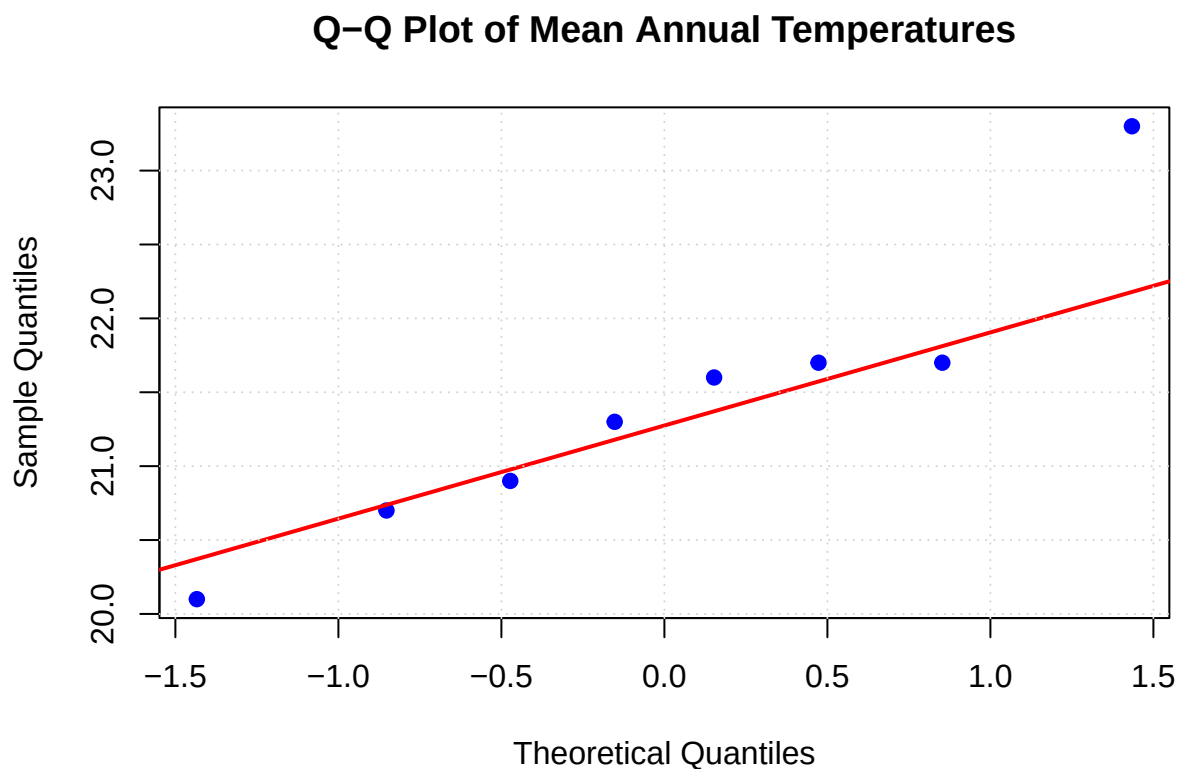
Now, we plot the data!!!

```

qqnorm(temperatures, main = "Q-Q Plot of Mean Annual Temperatures", pch = 19, col = "blue")
qqline(temperatures, col = "red", lwd = 2)

grid()

```



Since most of the data points lie on or around the red line, we confirm the assumption of normality.

Now, we construct a 95% confidence interval.

Let's use R to make this process easier...

```

variance_temp <- var(temperatures)

df <- length(temperatures) - 1
alpha <- 0.05
chi_sq_lower <- qchisq(alpha / 2, df, lower.tail = FALSE)
chi_sq_upper <- qchisq(1 - alpha / 2, df, lower.tail = FALSE)

variance_lower <- (df * variance_temp) / chi_sq_lower
variance_upper <- (df * variance_temp) / chi_sq_upper

sd_lower <- sqrt(variance_lower)
sd_upper <- sqrt(variance_upper)
cat("95% Confidence Interval for Standard Deviation:\n")

## 95% Confidence Interval for Standard Deviation:
cat("Lower Limit:", round(sd_lower, 3), "°C\n")

## Lower Limit: 0.626 °C
cat("Upper Limit:", round(sd_upper, 3), "°C\n")

## Upper Limit: 1.926 °C

```

Problem 8.4.1

Part A

Ultimately, we wish to solve the following to determine the confidence intervals:

$$\hat{p} \pm Z_{\alpha/2} \left(\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right)$$

, where $\hat{p} = \frac{x}{n}$. We will use R to speed up calculations...

```

x <- 412 # bush voters
n <- 768 # college grads
conf_level <- 0.95

p_hat <- x / n

z <- qnorm(1 - (1 - conf_level)/2)

SE <- sqrt(p_hat * (1 - p_hat) / n)

l_bound <- p_hat - z * SE
u_bound <- p_hat + z * SE

cat("We find the 95% Confidence Interval for p to be",
    round(l_bound, 5), "to", round(u_bound, 5))

## We find the 95% Confidence Interval for p to be 0.50119 to 0.57173

```

Part B

We find $\hat{p} = \frac{412}{768} = 0.5365$.

Therefore, we have:

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 0.0173$$

Using R,

```
z_lower <- qnorm(conf_level)
```

```
lower_conf_bound <- p_hat - z_lower * SE
```

```
cat("95% Lower Confidence Bound for p:", round(lower_conf_bound, 5))
```

```
## 95% Lower Confidence Bound for p: 0.50686
```

And so, we have arrived at our answer of 50.69% for the 95% LCB.